Human-Elicited vs. LLM-Generated Functional Requirements in a Smart Classroom System: A Paired Blind Study

Gary Alberto Sánchez Cornejo[0009-0009-9599-8806], Yeranick Esther Muñoz Quiñónez[0009-0005-7711-8730], and Winston Damián Cedeño Ávila[0009-0000-6086-6269]

Facultad de Ciencias de la Computación, Universidad Técnica Estatal de Quevedo, Quevedo, Los Ríos, Ecuador
gsanchezc6@uteq.edu.ec

Abstract. Context. Large Language Models (LLMs) are increasingly proposed as tools to support requirements elicitation, yet few studies compare their output against human-elicited requirements from the *same* real-world source material. **Objective.** We report a paired, blind quasi-experiment comparing the quality of functional requirements (FR) elicited by a human analyst team against FR generated by an LLM from the same corpus of ten anonymized stakeholder interviews, collected for a real smart-classroom management system (SIGA) deployed at a university in Ecuador. **Method.** Three independent judges, blind to the origin of each item, scored 51 requirements on five quality dimensions (completeness, absence of ambiguity, verifiability, source correctness, internal consistency) using a 1–5 Likert scale. Paired tests (t-test or Wilcoxon, chosen per dimension by Shapiro-Wilk normality) were applied with Holm-Bonferroni correction for the five comparisons; effect sizes (Cohen's d or Cliff's δ) were estimated with 95% bootstrap confidence intervals (10,000 replicates). **Results.** LLM-generated requirements scored numerically higher than human-elicited ones on all five dimensions, but no difference survived the Holm-Bonferroni correction ($p_{\mathrm{holm}} > 0.05$ for all dimensions). A post-hoc power calculation showed the three-judge panel reaches only 8.4% power to detect a medium effect ($d = 0.5$, $\alpha = 0.05$); at least 34 paired observations would be needed for 80% power. **Conclusions.** With the present sample size, we cannot confirm or rule out a quality difference between human- and LLM-elicited requirements; the study is substantially underpowered, and we report this limitation explicitly rather than over-interpreting the numerically favorable trend toward the LLM.

Keywords: Requirements elicitation · Large Language Models · Empirical study · Smart classrooms · Requirements quality · Latin America

1 Introduction

Requirements elicitation is one of the costliest and most error-prone phases of software development, and errors introduced here propagate downstream at a multiplicative cost [11]. SIGA (Sistema Inteligente de Gestión de Aulas) is a real, client-commissioned IoT and machine-learning platform for monitoring classroom occupancy, temperature, and equipment status at a public university in Ecuador; its requirements were elicited from ten stakeholder interviews (teaching staff, academic coordination, and facilities personnel) collected across two field rounds.

Large Language Models are increasingly proposed as elicitation aids, but most existing evidence is either qualitative or compares LLM output against a gold standard rather than against requirements produced by human analysts working from identical source material [3,2]. Cheng et al.'s systematic review of 238 studies found elicitation is one of the two most-studied RE phases for generative AI, yet identified reproducibility and hallucination as the two most frequently reported open challenges [3]. Closer to our design, Hymel and Johnson found LLM-generated requirements rated higher on alignment and completeness than human-expert output, with reviewers often misattributing the better output to the human expert [5]; Almeida et al. compared two LLMs generating requirements from transcribed stakeholder interviews and found both models struggled with ambiguity and requirement categorization despite good precision on non-functional requirements [1].

This paper contributes: (1) a paired, blind comparison of FR quality using the same interview corpus for both human and LLM elicitation, in a real, currently deployed system rather than a toy or textbook case; (2) a transparent, fully reproducible analysis pipeline (Sect. 4) that discloses two methodological complications encountered mid-study — the withdrawal of consent by one interview participant after the LLM had already processed their transcript, and an initial lack of inflection in the qualitative saturation curve, later traced to a one-to-one codebook and resolved by coding the remaining interviews — rather than silently absorbing them; (3) an explicit post-hoc power analysis that quantifies exactly how under-powered a small-panel blind evaluation is, as guidance for future replications.

RQ1. In which quality dimensions do functional requirements elicited by a human analyst team differ from those generated by an LLM from the same source material, in a real-world smart classroom management system?

RQ2. What statistical power does a three-judge blind panel achieve on such a comparison, and what panel size would a replication need?

2 Related Work

We conducted an abbreviated search (Google Scholar, arXiv, ACM DL; August 2026) using the terms *"LLM requirements elicitation"*, *"generative AI functional requirements evaluation"* and *"ChatGPT requirements quality"*, screening for peer-reviewed or preprint studies from 2023–2026 that generate or evaluate requirements-engineering artifacts with an LLM. Table 1 summarizes the ten most directly comparable studies found.

Table 1. Comparative summary of closely related work

Reference	Year	Design	Difference with our study
Cheng et al. [3]	2026	Systematic review (238 studies)	Maps the field; does not itself run a paired human-vs-LLM comparison
Arora et al. [2]	2024	Position/assessment paper	Discusses LLM roles in RE conceptually, no empirical panel evaluation
Vogelsang & Fischbach [13]	2024	Guideline paper	Methodological guidance for using LLMs in RE, not a comparative study
Lubos et al. [6]	2024	Case study	LLM assists QA of existing requirements; does not compare elicitation origin
Ronanki et al. [9]	2023	Exploratory case study	Explores ChatGPT assistance in elicitation; no blind judge panel
Ronanki et al. [10]	2023	Comparative study	Evaluates ChatGPT *as an evaluator* of user stories, not as a generator compared against humans
Görer & Aydemir [4]	2023	Tool paper	Generates interview scripts with LLMs; upstream of our elicitation step
Ray et al. [12]	2023	Case study	LLM-assisted requirement standardization, single-origin corpus
Hymel & Johnson [5]	2025	Blind rater comparison	Closest design match; different domain, no source-material identity guarantee, no pre-registration
Almeida et al. [1]	2025	Two-model case study	Compares two LLMs on the same interviews; does not include a human-elicited comparison arm

Research niche. No study we found combines all three of: (a) a paired design in which the human and LLM arms are elicited from the identical interview corpus, (b) a blind judge panel with a pre-registered analysis plan, and (c) a real, currently deployed system rather than a course exercise or synthetic case. This is the gap our study addresses, while explicitly disclosing the statistical power limits of a three-judge panel.

3 Methodology

3.1 Research questions and PICOC

See RQ1 and RQ2 above. The PICOC below frames RQ1; RQ2 is answered on the same design, taking the panel itself as the object of measurement. **Population:** functional requirements of the SIGA system (26 LLM-generated, 25 human-elicited). **Intervention:** LLM-assisted elicitation (single query, controlled temperature) from the full anonymized interview corpus. **Comparison:** the human analyst team's own elicitation process (interviews plus manual analysis) over the same corpus. **Outcome:** blind quality score (1–5) on five dimensions. **Context:** an academic Requirements Engineering course project for a real client system at a public Ecuadorian university.

3.2 Hypotheses

H_0: no significant difference in mean quality score between the human-elicited (Set B) and LLM-generated (Set A) requirements, for each of the five dimensions. H_1: a significant difference exists in at least one dimension.

3.3 Variables

Independent: requirement origin (Human / LLM). Dependent: 1–5 Likert score per dimension per judge. Control: identical source corpus for both arms; the same three judges score both sets, blind to origin, in a randomized order (fixed seed 20260802).

3.4 Design and participants

Paired quasi-experiment. Three judges, verifiably independent of the ten interviews that produced the source corpus, each scored all 51 blinded items using a standardized rubric. The project corpus comprises sixteen interviews; the ten cited here are those that were thematically coded and from which both requirement sets were derived (see T4). **Deviation from the pre-registered protocol:** the protocol recommended 3–5 judges; the study ran with the minimum of 3, which — as Sect. 4 quantifies — yields very low statistical power.

3.5 Instruments and reproducibility

The blind item package, scoring sheet, and judge instructions are included in the replication package (`06_Experimento/instrumentos/`). The full analysis pipeline (`06_Experimento/scripts_analisis/`, invoked via `make all`) regenerates every table and figure in Sect. 4 directly from the raw judge score sheets; no figure was produced manually.

3.6 Analysis plan and deviations from pre-registration

Per dimension: Shapiro-Wilk normality test on each arm's per-judge means; paired t-test if both arms are normal, Wilcoxon signed-rank test otherwise; Holm-Bonferroni correction across the five dimensions; effect size (Cohen's d or Cliff's δ) with a 95% bootstrap confidence interval (10,000 resamples, seed 20260802). The study protocol was pre-registered on OSF (`https://osf.io/7pq3h`). Three deviations from that pre-registration are declared explicitly, in line with open-science reporting norms [8]: (1) the Holm-Bonferroni correction, the bootstrap confidence intervals, and the Levene homogeneity-of-variance test were added after registration, none of them were in the original analysis plan, and all three make the analysis *more*, not less, conservative; (2) one source interview (coded EV-15) was excluded from the human-elicited evidence base after the participant withdrew informed consent — this occurred *after* the LLM had already processed that transcript to generate Set A and after the blind judging had already taken place, so Set A's source material and Set B's evidentiary support are no longer perfectly symmetric; re-running the full experiment was not feasible within the project timeline, and this asymmetry is carried forward as a construct-validity threat (Sect. 6); (3) the qualitative saturation curve did not show an inflection point while only ten interviews were coded, which is why a post-hoc power calculation was introduced as the alternative adequacy criterion, following standard practice for underpowered small-N designs [7]; completing the coding of all sixteen interviews later satisfied the saturation criterion directly (Sect. 4), and the power calculation is retained because it answers RQ2 rather than because it is needed as a substitute.

4 Results

4.1 RQ1: Quality differences between human- and LLM-elicited requirements

Table 2 reports descriptive statistics. On every one of the five dimensions, the LLM arm's median and mean score were equal to or higher than the human arm's (e.g. source correctness: median 5.000 vs. 4.000; mean 4.487 vs. 4.293).

Table 3 reports the normality (Shapiro-Wilk) and homogeneity-of-variance (Levene) tests that determined which paired test was applied per dimension.

Table 4 reports the paired hypothesis tests with Holm-Bonferroni correction and the corresponding effect sizes with 95% bootstrap confidence intervals. Before correction, *Consistencia_interna* showed a nominally significant difference ($t = -9.123$, $p = 0.012$); after Holm-Bonferroni correction across the five dimensions, this rises to $p_{\text{holm}} = 0.059$ and is **no longer significant** at $\alpha = 0.05$. **No dimension survives correction.**

Effect sizes are large in magnitude (Cohen's d up to -5.267 for *Consistencia_interna*, all favoring the LLM arm) but their 95% bootstrap confidence intervals are extremely wide (e.g. $[-42.72, 0.00]$), a direct consequence of bootstrapping from only three paired observations. Figure 1 visualizes this.

Table 2. Descriptive statistics by quality dimension and requirement source

Dimension	Source	Median	Mean	SD	Min	Max	IQR
Completitud	Humano	4.000	3.907	1.016	2.000	5.000	2.000
Completitud	LLM	4.000	4.115	0.868	2.000	5.000	1.000
Ausencia_ambiguedad	Humano	4.000	3.613	1.051	1.000	5.000	1.000
Ausencia_ambiguedad	LLM	4.000	3.859	0.977	1.000	5.000	2.000
Verificabilidad	Humano	4.000	3.653	1.033	1.000	5.000	1.500
Verificabilidad	LLM	4.000	3.872	1.121	1.000	5.000	2.000
Correccion_fuente	Humano	4.000	4.293	0.731	3.000	5.000	1.000
Correccion_fuente	LLM	5.000	4.487	0.597	3.000	5.000	1.000
Consistencia_interna	Humano	4.000	4.147	0.817	2.000	5.000	1.000
Consistencia_interna	LLM	4.000	4.385	0.669	3.000	5.000	1.000

Table 3. Assumption checks: normality (Shapiro-Wilk) and homogeneity of variances (Levene)

Dimension	n	SW p (Human)	SW p (LLM)	Normal (H)	Normal (L)	Levene W	Levene p	Equal var.
Completitud	3	0.780	0.537	Sí	Sí	0.122	0.745	Sí
Ausencia_ambiguedad	3	0.253	0.843	Sí	Sí	0.002	0.969	Sí
Verificabilidad	3	0.780	0.637	Sí	Sí	0.458	0.536	Sí
Correccion_fuente	3	0.298	0.000	Sí	No	0.115	0.751	Sí
Consistencia_interna	3	0.843	0.537	Sí	Sí	0.002	0.964	Sí

Table 4. Paired hypothesis tests by dimension, with Holm-Bonferroni correction and effect sizes

Dimension	Test	Stat.	Value	p	p_{holm}	Effect size	Value	95% CI
Completitud	t apareada	t	-1.804	0.213	0.426	Cohen d (apareado)	-1.042	[-2.117, 0.000]
Ausencia_ambiguedad	t apareada	t	-4.071	0.055	0.222	Cohen d (apareado)	-2.350	[-18.384, 0.000]
Verificabilidad	t apareada	t	-2.534	0.127	0.380	Cohen d (apareado)	-1.463	[-12.958, 0.000]
Correccion_fuente	Wilcoxon de rangos con signo	W	1.000	0.500	0.500	Cliff delta	-0.556	[-1.000, 1.000]
Consistencia_interna	t apareada	t	-9.123	0.012	0.059	Cohen d (apareado)	-5.267	[-42.724, 0.000]

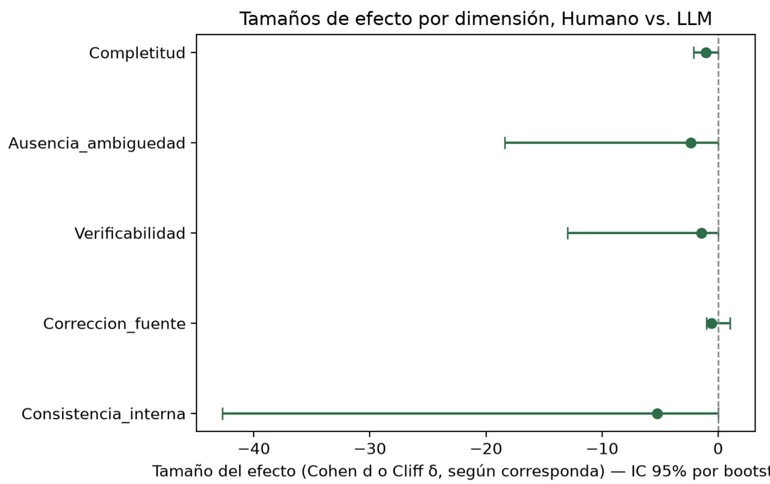

Fig. 1. Effect sizes by dimension with 95% bootstrap confidence intervals ($n = 3$ judges). All point estimates favor the LLM arm, but every interval crosses zero.

4.2 Inter-rater agreement

Table 5 reports pairwise Cohen's κ (linear weights) and Fleiss' κ across the three judges. Agreement is *fair* by conventional benchmarks (Fleiss' κ between 0.294 and 0.340 across dimensions), consistent with the inherent subjectivity of requirements-quality judgments and a further reason for caution in interpreting the point estimates above.

Table 5. Inter-rater agreement: pairwise Cohen kappa and overall Fleiss kappa

Dimension	κ J1–J2	κ J1–J3	κ J2–J3	Fleiss κ
Completitud	0.474	0.646	0.420	0.338
Ausencia_ambiguedad	0.654	0.549	0.494	0.340
Verificabilidad	0.646	0.550	0.419	0.296
Correccion_fuente	0.444	0.396	0.316	0.294
Consistencia_interna	0.577	0.467	0.294	0.333

4.3 RQ2: Statistical power of a three-judge blind panel

Figure 2 shows the thematic-code saturation curve. An earlier version of this analysis, over the ten interviews coded at the time, reached **no inflection**: 36 distinct codes, and a mean of 3.667 new codes in the last three interviews against a 5% threshold of 1.8. Post-hoc inspection attributed this to the codebook rather than to the corpus — each of the 36 codes applied to exactly one fragment, with no axial pass merging codes standing for the same underlying theme, and with a strictly one-to-one codebook the curve cannot flatten at any interview count.

That diagnosis was acted upon rather than left standing. The six interviews of the terminal round were coded on 6 September 2026 by the three authors, splitting whole interviews and working against the existing codebook, and the structural obstacle disappears: the corpus now holds **136 coded fragments under 50 codes**, 2.72 fragments per code against the previous 1.00, and singleton codes fall from 100% to 44%. The mean number of new codes in the last three interviews is **1.333** against a threshold of 2.50, so the criterion is met, first at the fourteenth interview and then at the fifteenth and sixteenth.

Two caveats. The result does not depend on ordering: the six closing interviews ran on a single day, so their position is set by evidence number rather than clock time, and the criterion holds across **all 720 permutations** of that block (worst case 2.333 against the 2.50 threshold). Against us, the coding instructions explicitly asked coders to reuse an existing code wherever one fitted — methodologically right, since proliferating synonyms distorts the curve the other way, but it pushes toward the outcome obtained, and the coders' reuse rates (59%, 63%, 64%) show only that the tendency was evenly spread, not absent.

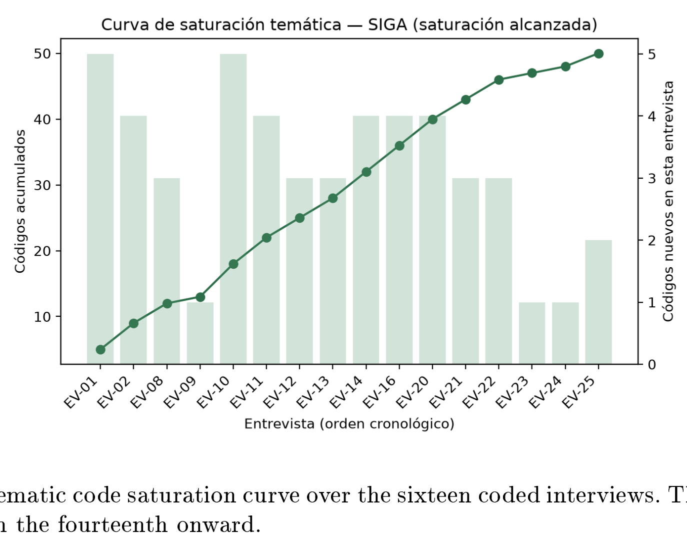

Fig. 2. Thematic code saturation curve over the sixteen coded interviews. The criterion is met from the fourteenth onward.

As the alternative adequacy criterion, Table 6 reports a post-hoc power calculation for the paired-judge design: to detect a medium effect ($d = 0.5$) at $\alpha = 0.05$ with 80% power, **34** paired observations would be required; the present $n = 3$ achieves only **8.4%** power.

Table 6. Statistical power calculation for the paired t-test

Target d	α	Target power	n needed	n needed (int.)	n actual	Power achieved
0.5000	0.0500	0.8000	33.3700	34	3	0.0841

4.4 Sensitivity analysis: the requirement as the unit of analysis

The pre-registered plan treats *the judge* as the unit of analysis, averaging each judge's scores across items within each origin and applying a paired test over $n = 3$. That choice controls for judge severity, but it discards most of the information collected — 765 individual ratings reduced to three paired means — and yields effect-size confidence intervals too wide to interpret.

We therefore report an **exploratory, post-registration** sensitivity analysis that takes *the requirement* as the unit: the three judges' scores are averaged within each item, giving 25 human-elicited and 26 LLM-generated observations per dimension. Because the two sets contain different requirements rather than the same requirement measured twice, the comparison is between independent samples. Normality is checked per group with Shapiro-Wilk; Welch's t-test is used when it is not rejected and the Mann-Whitney U test otherwise, with Holm-Bonferroni correction over the five dimensions. Effect sizes are Hedges' g or Cliff's δ as appropriate, with 95% bootstrap confidence intervals (10,000 replicates, seed 20260802). This analysis is declared in the deviation log; it **supplements and does not replace** the pre-registered analysis of Sect. 4.

Table 7. Sensitivity analysis with the requirement as the unit of analysis (exploratory, post-registration). Human $n = 25$, LLM $n = 26$; judge scores averaged per item.

Dimension	Human	LLM	Test statistic	p (Holm)	Effect size [95% CI]
Completeness	3.91	4.12	287.000	1.0000	-0.117 [-0.434, 0.200]
Non-ambiguity	3.61	3.86	-0.931	1.0000	-0.257 [-0.824, 0.298]
Verifiability	3.65	3.87	278.500	1.0000	-0.143 [-0.459, 0.177]
Source correctness	4.29	4.49	267.000	1.0000	-0.178 [-0.477, 0.128]
Internal consistency	4.15	4.38	265.000	1.0000	-0.185 [-0.478, 0.126]

Three observations follow. First, the achieved power for a medium effect ($d = 0.5$) rises from **8.4%** to **41.7%**; still below the conventional 80% threshold, but a fivefold improvement obtained from data already collected. Second, the effect sizes become interpretable: Cliff's δ ranges from -0.117 to -0.185 and Hedges' g is -0.257, all small, all favouring the LLM set slightly, and **all with confidence intervals crossing zero**. Third, and most importantly, **both analyses reach the same conclusion**: after correction for multiple comparisons, no dimension shows a statistically significant difference between human-elicited and LLM-generated functional requirements.

That agreement is the point of a sensitivity analysis. The pre-registered result was not significant and neither is this one, so the conclusion does not depend on the analytical choice — which is precisely what a reader should be able to check before trusting a null result obtained from a small panel.

5 Discussion

5.1 Implications for research

Every point estimate in this study favors the LLM arm, echoing Hymel and Johnson's finding that LLM-generated requirements can be rated at least as favorably as human-expert output [5]. However, our study demonstrates concretely *how misleading* such a favorable trend can be without an accompanying power analysis: a $d = -5.267$ point estimate coexists with a 95% CI of $[-42.72, 0.00]$ and a corrected p of 0.059. We recommend that future paired blind LLM-vs-human RE studies report power calculations alongside effect sizes as standard practice, not as an afterthought.

5.2 Implications for practice

For practitioners considering LLM-assisted elicitation, our descriptive results (higher LLM medians across all five dimensions) are consistent with LLMs being a *usable first-pass generator* subject to human review, echoing Lubos et al.'s use of LLMs for requirements QA rather than autonomous generation [6]. We do not have statistical grounds to recommend replacing human elicitation.

5.3 Surprising or counterintuitive findings

The dimension with the strongest raw effect (*Consistencia_interna*) is also the one where the correction changes the substantive conclusion most (from "significant" to "not significant"). This is a useful illustration, for students and practitioners new to empirical SE research, of why multiple-comparison correction is not merely a formality.

5.4 Lessons learned

For undergraduate teams running similarly small empirical studies: (1) a pre-registered protocol should include the maximum feasible number of judges from the outset — retrofitting a power calculation after data collection, as we did, can only diagnose the problem, not fix it; (2) any participant exclusion after data has already propagated downstream (here, into the LLM's source material) should be documented as a first-class deviation, not silently patched; (3) a codebook intended to demonstrate saturation must include an explicit axial-coding merge step from the start — a codebook in which every code applies to exactly one fragment cannot produce an inflection at any sample size, and no amount of further interviewing will fix it.

6 Threats to Validity

We report two threats per category following Wohlin et al.'s classification. For each we state the mitigation actually applied and the limitation that remains after mitigating, rather than closing the discussion at the mitigation.

6.1 Internal validity

T1. Three-judge panel. With three judges the design reaches 8.4% power to detect a medium effect (Table 6); a true difference would most likely go undetected. *Mitigation:* we compute and report the power explicitly, and we add a pre-specified-deviation sensitivity analysis that takes the requirement rather than the judge as the unit of analysis, raising power to 41.7% (Sect. 4.4). *Remaining limitation:* 41.7% is still half the conventional 80% threshold, so a negative result here cannot be read as evidence of equivalence.

T2. Fair inter-rater agreement. Fleiss' $\kappa \approx 0.3$ (Table 5) indicates the judges applied the rubric with only fair consistency, adding measurement noise on top of the small sample. *Mitigation:* evaluation was blind, the de-blinding key is deposited with the replication package, and the agreement statistic is reported rather than omitted. We further attempted to widen the panel from three to ten raters on two separate occasions, and both attempts are deposited in full (`06_Experimento/panel_ampliado/`). *Remaining limitation:* neither attempt produced usable agreement, and the failure is informative about the instrument rather than about the object of study. With ten raters, Fleiss' κ fell to -0.008 in the first round and -0.026 in the second, despite the second round shortening the instrument from 255 to 108 judgements per rater, dropping the one dimension that could not be answered from the materials supplied, printing the rubric anchors on the response sheet itself, and running a group calibration beforehand. This is not an artefact of a concentrated response scale: observed exact pairwise agreement was 37.5% against the 39.1% expected by chance under the raters' own marginals, and no subset of raters, ordered by internal consistency, yields a positive κ. The second round embedded three undisclosed repeated statements, and these locate the noise: three of the seven raters contradicted their own earlier score of the identical statement by up to three points on a five-point scale within the same sitting, and one reproduced none of his or her twelve prior scores. The variance is therefore *intra*-rater rather than inter-rater, which means it cannot be reduced by recruiting more raters. We consequently retain the three pre-registered judges as the primary analysis — a decision rule fixed in writing before the second round was convened, which named no numeric threshold — and we report the extended panel as evidence that this rubric requires trained evaluators. A replication that recruits an untrained student panel to gain statistical power should expect to lose measurement stability instead.

6.2 External validity

T3. Single setting. Findings come from one system, one institutional domain, one LLM configuration and Spanish-language transcripts from Ecuador. *Mitigation:* the instrument, the prompts, the labelled corpus and the anonymised transcripts are deposited under CC BY 4.0, so the study can be repeated in another setting without contacting the authors. *Remaining limitation:* no replication has been carried out, so transferability is an argument about available materials and not an empirical claim.

T4. The compared sets rest on a narrower corpus than the coded one. Elicitation for the compared sets closed at ten valid interviews. Field work later reopened: a terminal round conducted on 3 September 2026 added six teacher interviews (EV-20 to EV-25), bringing the project corpus to sixteen, and those six were thematically coded on 6 September 2026. The codebook and the saturation curve therefore now cover all sixteen. *The two compared requirement sets do not.* Both were generated from the ten-interview material and were not regenerated, so the quasi-experiment's unit of comparison is still anchored to ten interviews while the qualitative analysis around it is anchored to sixteen. *Mitigation:* the two figures are kept distinct wherever the corpus is described, and the source material handed to the model is deposited verbatim, so a reader can check which interviews fed each set; the closure at ten is declared in writing with its date and a signed record of the instruction under which the field was closed. *Remaining limitation:* regenerating both sets from the sixteen-interview corpus would change what is compared and would need re-registering, so it is future work rather than a deadline fix. The six added interviews are all of the teaching profile, so the corpus grew in depth for one role rather than in breadth: a requirement only a non-sampled role would have raised still cannot appear in either set, affecting both arms equally.

6.3 Construct validity

T5. Prior exposure of the model. The LLM had partial prior exposure to Set B within the same session used to generate Set A, documented in `06_Experimento/prompts_llm/prompt_ll`
Mitigation: the exposure is recorded verbatim in the prompt file and declared as a deviation from the registered protocol rather than being described after the fact. *Remaining limitation:* the contamination cannot be undone without regenerating Set A in a clean session, which is listed as future work; until then, any advantage measured for the LLM may be partly an artefact of having seen the comparison set.

T6. Non-identical source corpora. The corpus fed to the LLM comprises eleven interviews, while the corpus backing the human-elicited set comprises ten after EV-15 was excluded for withdrawal of consent (Sect. 3.6). *Mitigation:* the mismatch is quantified and declared, and the excluded material was removed from the deposited package rather than silently retained. *Remaining limitation:* the paired design assumes a shared source, and that assumption is therefore satisfied only approximately; the pairing remains valid at the item level but not at the corpus level.

6.4 Conclusion validity

T7. Wide interval estimates. Bootstrap confidence intervals for every effect size cross zero, so the direction of the effect is not established. *Mitigation:* we report intervals alongside point estimates and refuse to interpret the numerically favourable trend as a finding. *Remaining limitation:* with this sample the study cannot distinguish absence of an effect from an effect it lacks the power to detect, and we state the result as inconclusive rather than negative.

T8. Multiple comparisons. Testing five quality dimensions inflates the family-wise error rate; before correction, internal consistency reached $p = 0.012$ and would have been reported as significant. *Mitigation:* Holm–Bonferroni correction is applied and both the uncorrected and the corrected p values are reported, so the reader can see what the correction changed. *Remaining limitation:* the correction lowers power further, compounding T1; with five dimensions and this sample size, only a large effect could have survived it.

7 Conclusions and Future Work

RQ2. A three-judge blind panel achieves 8.4% power for a medium effect ($d = 0.5$, $\alpha = 0.05$); 34 paired observations would be required to reach the conventional 80%. A replication should therefore either enlarge the panel or treat the requirement, rather than the judge, as the unit of analysis, as the sensitivity analysis in Sect. 4 illustrates. We report this so that the next study does not repeat an under-powered design without knowing it.

RQ1. Across all five quality dimensions, LLM-generated functional requirements scored numerically at or above human-elicited requirements, but after Holm-Bonferroni correction **no difference is statistically significant**, and a post-hoc power calculation shows the three-judge design is highly underpowered (8.4% vs. a conventional 80% target). We therefore conclude that this study cannot confirm a quality difference between the two elicitation approaches, and explicitly decline to over-interpret the favorable point estimates.

Contributions. (1) A fully reproducible paired-blind comparison pipeline, released with the replication package; (2) transparent disclosure of a mid-study consent withdrawal and its downstream effect on corpus symmetry; (3) a concrete, quantified illustration of how multiple-comparison correction and power analysis change substantive conclusions in small-panel RE evaluations.

Future work. We recommend, in order of priority for anyone extending this dataset: (1) expanding the judge panel to at least 34 to reach conventional power, using the anonymized replication package as source material; (2) re-generating both requirement sets from the full sixteen-interview corpus, which would restore paired symmetry and align the compared sets with the coded corpus, and which needs to be re-registered because it changes what is compared; (3) widening the terminal round beyond the teaching profile, since saturation was reached with a final stretch drawn from a single role.

Data and materials availability

The replication package supporting this study is openly available in Zenodo under the Creative Commons Attribution 4.0 International licence (CC BY 4.0): `https://doi.org/10.5281/zenodo.22663649`. The study protocol was pre-registered in the Open Science Framework: `https://doi.org/10.17605/OSF.IO/7PQ3H`, with deviations documented in `06_Experimento/registro_previo/`. The source repository is permanently archived in Software Heritage under the identifier `swh:1:snp:861295fead33417e3efc2753fd4a34897014a891`. Analysis code is released under the Apache-2.0 licence.

Use of AI-assisted technologies

A Large Language Model was used in two clearly separated capacities. First, as the *object of study*: Set A of functional requirements was generated under controlled conditions, with the exact model, version, temperature, top-p, seed and query timestamp recorded in the replication package (`06_Experimento/prompts_llm/`). Second, as *writing support*: an AI coding assistant (Claude, Anthropic) was used to build and debug the reproducible analysis pipeline (`06_Experimento/scripts_analisis/`) under the direct supervision of the authors, who verified every statistical method against the methodology declared in the pre-registered protocol; to polish the wording of paragraphs whose content and data interpretation were produced by the authors; and to restructure the paper so that the second research question, already answered by analyses the authors had conducted, is stated explicitly. No result, figure, table, conclusion or bibliographic reference in this manuscript was produced by a Large Language Model; every reference was individually verified against its publisher or preprint page and every DOI/arXiv identifier resolved before submission. The authors take full responsibility for the content of this work.

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